Physics Worksheet

Integral Practice

Name: ____________ **Student #:** ____________ **Student Email #:** ________

1. **Integration**
   Compute:
   $$\int 3x^2[x^3+5]^4 dx$$

2. **Modelling and Integrating**
   Alex is a long distance crawler. Their coach, James, who happens to be a physicist is attempting to model their crawling as a function of time. He finds that their jerk($\frac{d^3}{dt^3}x$) can be modelled as $j(t) = te^{-t}$ in $\mathrm{km\,h^{-3}}$ based on data from their races.

   (a) Using this, find $v(t)$in $\mathrm{km\,h^{-1}}$.

   (b) How far can alex crawl in 3 hours? Assume initial velocity, acceleration, and jerk are all 0.

   (c) James wants to find the point where Alex runs out of energy and stops crawling. How long can Alex crawl for until they stop making substantial progress?